

Proposal: Evaluating Modern x86 CPU Architectures for Numerical Computing in Physics Based Simulations

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Background

As computational physics and numerical simulation become increasingly central to scientific research and engineering, the choice of processor architecture has a meaningful impact on the time it takes to reach a solution and researcher productivity. Two dominant, but fundamentally different design philosophies now compete in the consumer market for x86 CPU architectures. Intel's hybrid Performance-core/Efficient-core (P-core/E-core) architecture and AMD's 3D V-Cache technology, which vertically stacks additional L3 cache directly onto the compute die.

Intel introduced its hybrid architecture with the 12th Generation "Alder Lake" processors in 2021, combining high clock speed P-cores optimized for single threaded performance with smaller, power efficient E-cores designed for background and multithreaded workloads. Intel's Thread Director dynamically assigns tasks to the appropriate core type at the hardware level (Intel, 2024). The Intel Core i9-14900K, part of the 14th Generation "Raptor Lake Refresh" lineup, features 8 P-cores and 16 E-cores for a total of 24 cores and 32 threads, representing the most mature iteration of this hybrid design.

AMD has taken a different approach. Rather than mixing core types, AMD's 3D V-Cache technology uses TSMC's hybrid bonding process to stack additional SRAM vertically on the compute die, tripling the available L3 cache without increasing die area (AMD, 2022). The AMD Ryzen 9 9800X3D, built on the "Zen 5" architecture with 2nd Generation 3D V-Cache, features 8 homogeneous cores with 96 MB of L3 cache. AMD has demonstrated that this enlarged cache delivers significant performance gains in cache sensitive workloads, including computational fluid dynamics (CFD) and finite element analysis (FEA) simulations (AMD, 2023).

While extensive benchmarking exists for gaming and general productivity, there is a gap in the literature regarding how these two architectural approaches perform on numerical computing

workloads, specifically dense linear algebra, Monte Carlo simulation, and GPU accelerated physics solvers. Most existing comparisons use synthetic benchmarks or gaming frame rates rather than scientific computing metrics such as FLOPS, memory bandwidth utilization, and cache hit rates. The LINPACK benchmark, originally developed by Dongarra (2003) and used to rank the world's fastest supercomputers in the TOP500 list, provides a well established methodology for measuring floating point performance, but it is rarely applied to compare consumer desktop CPUs in the context of scientific workloads.

This study aims to fill that gap by directly comparing the Intel Core i9-14900K and the AMD Ryzen 9 9800X3D across a suite of numerical and physics based workloads, using an identical NVIDIA RTX 4070 GPU, and matched system configurations to isolate CPU architectural differences.

Methodology

This project will employ a controlled experimental design in which only the CPU and its associated motherboard differ between test configurations. All other components, NVIDIA GEFORCE RTX 4070 GPU, 32GBs of TeamForce Delta DDR5 6400MHZ RAM, 2TB M.2 NVMe SSD, Windows 11 Home Edition, and a Artic Freezer III Pro AIO cooling solution, will remain constant across both test platforms. The two motherboards will be selected from the same chipset tier (Intel Z790 and AMD B650) and matched as closely as possible in terms of VRM quality, memory topology, and PCIe lane configuration, so as to minimize performance differences attributable to the motherboard rather than the CPU.

The benchmarking suite will consist of the following workloads, chosen to stress different aspects of CPU architecture:

1. **Dense Linear Algebra (LINPACK):** The High Performance Linpack (HPL) benchmark will measure peak double precision floating point throughput (GFLOPS). This workload is compute bound and will primarily exercise raw floating point execution units and memory bandwidth.
2. **Monte Carlo Simulation:** A custom Monte Carlo simulation estimating π and solving

stochastic differential equations will test multithreaded random number generation and floating point throughput across all available cores. This will reveal how Intel’s heterogeneous core layout (P+E) compares to AMD’s homogeneous 8-core design under heavy parallel workloads.

3. **N-Body Gravitational Simulation:** An N-body particle simulation using both CPU only and GPU accelerated (CUDA) code paths will measure how each CPU performs when feeding work to the RTX 4070. This tests the CPU by observing GPU data transfer efficiency and whether one architecture introduces bottlenecks in GPU utilization.
4. **Finite Element Analysis (FEA):** Using the open source FEA solver CalculiX, a structural mechanics problem of fixed complexity will be solved on each platform, measuring wall clock time and memory bandwidth utilization.
5. **Matrix Multiplication and FFT:** Large dense matrix multiplications (via OpenBLAS) and Fast Fourier Transforms (via FFTW) will isolate cache performance, directly testing whether AMD’s 96 MB L3 cache provides a measurable advantage over Intel’s 36 MB L3 cache for problems whose working sets fit within the larger cache but not the smaller one.

Each benchmark will be run a minimum of five times per configuration, and results will be reported as means with standard deviations. Hardware performance counters will be used to collect cache hit/miss rates, instructions per cycle (IPC), and memory bandwidth utilization to provide architectural insight beyond raw throughput numbers. CPU power consumption will be measured by using HWiNFO software.

Anticipated Outcomes

This research is expected to produce the following results:

A quantitative comparison of Intel hybrid core and AMD 3D V-Cache architectures across five distinct numerical workload categories, reported in standardized metrics (GFLOPS, wall clock time, cache hit rates, performance-per-watt). Based on existing literature showing significant cache sensitivity in FEA and CFD workloads at the server level (AMD, 2023), I hypothesize that

the Ryzen 9 9800X3D will outperform the i9-14900K on cache sensitive workloads (FEA, FFT, and matrix operations), while the i9-14900K may hold an advantage on heavy parallel workloads (Monte Carlo) due to its higher thread count. The N-body GPU-accelerated test will reveal whether either platform introduces a CPU side bottleneck when paired with an RTX 4070 GPU.

A final written report will present all data, statistical analysis, and architectural explanations for observed performance differences. The report will include recommendations for students and researchers selecting hardware for numerical computing on a budget.

Significance and Project Benefits

This project contributes to the understanding of how modern consumer CPU architectures, hybrid core designs and 3D stacked cache, translate to realworld scientific computing performance. While server class benchmarks from AMD and Intel exist, they use enterprise hardware costing thousands of dollars. This study provides data relevant to students, independent researchers, and small labs who purchase consumer desktop hardware for computational work.

The intended audience includes undergraduate and graduate students in physics, engineering, and computer science who use desktop workstations for simulation and numerical analysis, as well as faculty advisors making hardware procurement decisions for teaching and research labs.

The new knowledge gained will be a direct, controlled comparison of two fundamentally different x86 design strategies on scientific workloads, something not currently available in the academic literature or the consumer hardware review space, which focuses almost exclusively on gaming and content creation.

References

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Timeline

Dates	Tasks
Jan 1 - Jan 15, 2027	Procure AMD Ryzen 9 9800X3D and B650 motherboard; assemble second test configuration; verify hardware stability
Jan 16 - Jan 31, 2027	Install operating systems (Windows 11, Ubuntu 24.04); install and configure benchmarking software (HPL, CalculiX, OpenBLAS, FFTW, CUDA toolkit); develop custom Monte Carlo and N-body simulation code
Feb 1 - Feb 28, 2027	Execute full benchmark suite on Intel i9-14900K configuration; collect performance counter data and power measurements; record at least 5 runs per benchmark
Mar 1 - Mar 31, 2027	Swap RTX 4070 and RAM to AMD configuration; execute identical benchmark suite on Ryzen 9 9800X3D; collect matching data
Apr 1 - Apr 15, 2027	Perform statistical analysis; generate comparison charts and tables; identify architectural explanations for observed differences using cache/IPC data
Apr 16 - Apr 30, 2027	Write final report; prepare presentation materials for UCF Student Scholar Symposium

Budget

Item	Unit Cost	Total
AMD Ryzen 9 9800X3D processor	1 - \$480.00	\$480.00
ASUS Prime B650M-A II	1 - \$129.99	\$129.99
Thermal paste (Noctua NT-H1, 3.5g)	2 - \$9.90	\$19.80
Total		\$629.79

Budget Justification: The AMD Ryzen 7 9800X3D and a compatible B650 motherboard are required to create the second test platform. All other components (RTX 4070 GPU, DDR5 RAM, NVMe SSD, CPU cooler, power supply, and case) are already owned by the researcher and will be shared between configurations. Thermal paste is needed for safe CPU installation and removal. CPU power consumption will be monitored using HWiNFO, an open source software tool, at no cost. All benchmarking software used in this project is open source and free of charge. Total requested funding: \$629.79.